



USL Assignment 3

Implement the Linear Discriminant Analysis algorithm

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LDA THEORY

1. Introduction to Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is a supervised dimensionality reduction and classification technique used to transform a dataset into a lower-dimensional space while preserving maximum class separability.

Unlike Principal Component Analysis (PCA), which focuses only on variance, LDA explicitly considers class labels and attempts to find feature combinations that best separate different classes.



2. Mathematical Concept Behind LDA

Given a dataset with multiple features belonging to different classes:

1. The mean vector of each class is computed
 2. The within-class scatter matrix is calculated to measure spread inside each class
 3. The between-class scatter matrix is calculated to measure separation between class means
 4. Eigenvalues and eigenvectors are computed from the scatter matrices
- Eigenvectors represent the discriminant directions
 - Eigenvalues indicate discriminating power

The eigenvectors corresponding to the largest eigenvalues form the reduced discriminant space.

3. Steps of LDA Algorithm

- Standardize the dataset
- Compute mean vectors for each class
- Compute within-class scatter matrix
- Compute between-class scatter matrix
- Solve eigenvalue problem
- Select top discriminant vectors
- Transform dataset into reduced discriminant space



4. Example of LDA

Consider a dataset containing student information:

- study hours
- attendance percentage

Students are categorized into Pass and Fail.

LDA finds a projection direction that maximizes separation between the two groups while minimizing variation within each group. The dataset is then projected onto this direction, making classification simpler and more accurate.

5. LDA Applied to Glass Identification Dataset

- The Glass Identification dataset consists of multiple chemical composition measurements describing glass samples. These chemical attributes collectively determine the type of glass.
- LDA was applied to:
 - identify the most discriminative chemical combinations
 - reduce dimensionality of the dataset
 - improve class separability before classification
- The transformed dataset was then used to train a classification model and compare accuracy before and after applying LDA.



6. Inferences from LDA Output

After applying LDA to the Glass Identification dataset, the explained variance ratio indicated how much discriminative information was captured by each linear discriminant component.

The projection of the dataset onto the first two discriminant components showed visible grouping of several glass types, confirming that chemical composition strongly influences classification.

The classification accuracy obtained after applying LDA was comparable to or slightly better than the accuracy obtained without dimensionality reduction. This indicates that LDA successfully reduced feature dimensionality while preserving essential class-separating information.

7. References

https://scikit-learn.org/stable/modules/generated/sklearn.discriminant_analysis.LinearDiscriminantAnalysis.html

<https://archive.ics.uci.edu>